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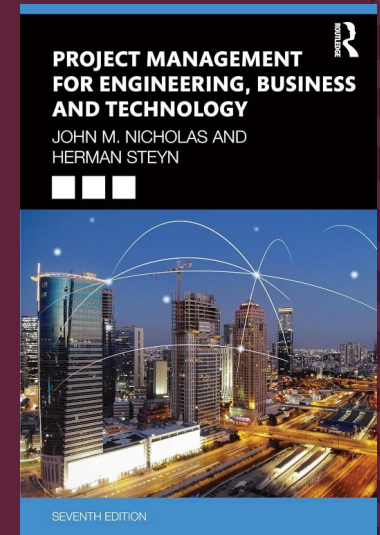
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CHAPTER 5 & 13: PROJECT EXECUTION, MONITORING, AND CONTROLLING

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Adapted from slides
prepared by

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Herman Steyn*



Chapter Mapping



Chapter 5

- 5.1 Phase C – Execution
- 5.2 Detail design stage
- 5.3 Production / Build Stage

Chapter 13

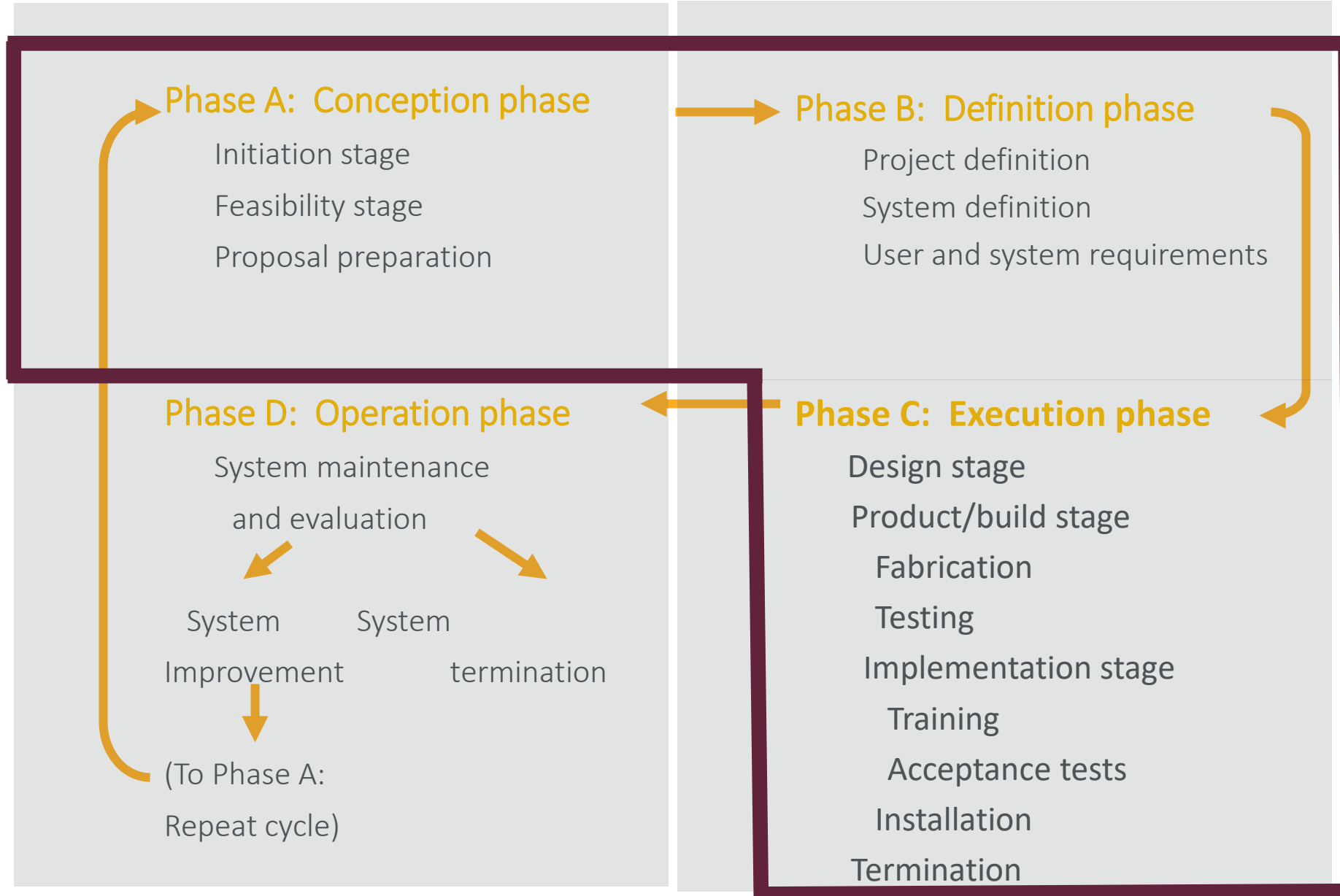
- 13.3 Monitoring and control process
- 13.4 Control emphasis
- 13.5 Work packages and control accounts

Introduction



- Up to now, **Conception** and **Project Definition** phases of the project life cycle
- Now we deal with what happens after we have planned and defined the project, **Project Execution**
- We know that no plan works out perfectly and we need to keep the project on target, track progress and assist with obstacles, **Project Control**

Project Life Span: Execution



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5.1 Project Execution

Execution Phase typically includes these stages:



Detail design



Production



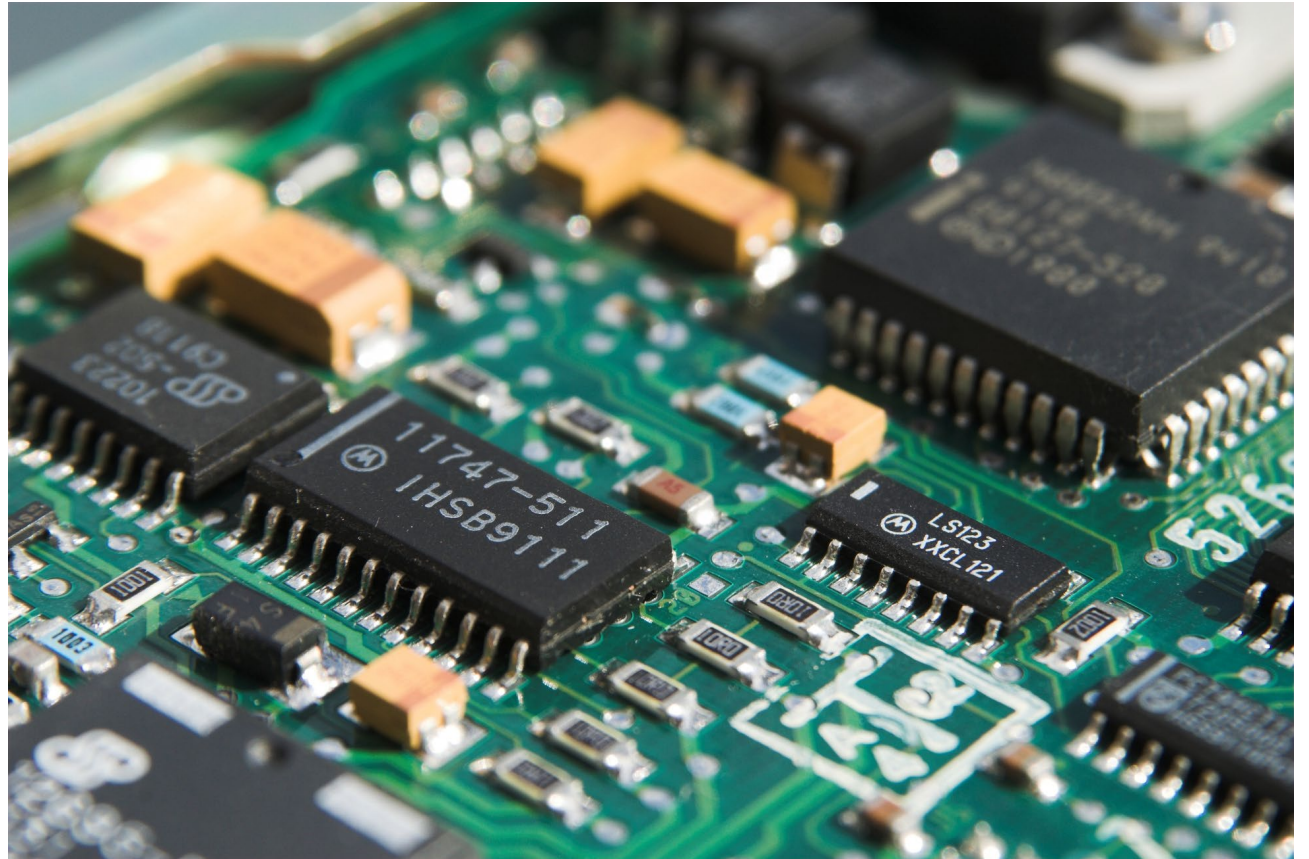
Implementation

5.1 Project Execution

Project Executions differ, depending on the project purpose

- Example: Hardware development typically has the following stages:

Design, Development and Production



5.1 Project Execution

Project Executions differ,
depending on the project purpose

Example: **Construction** typically has the
following stages:

Design and Build



5.1 Project Execution



Project Executions differ, depending on the project purpose

- That is why organisations have **customised project methodologies**; their own unique project phases and stages
- We will look at the stages of **Design** and **Production/build**

5.2 Detail Design Stage



- In this stage, specifications are converted to plans, sketches and drawings
- The output of this stage are:
 - Blueprints, flow charts, schematic diagrams, showing
 - The end-item's components, dimensions relationships and configuration
- In the process of detail design, we take a **breakdown approach**
 - Tiers, subsystems, components, parts (WBS, and other topics covered earlier)

5.2 Detail Design Stage



- Planning for Production / Build and later stages:
 - This entails designing not only to satisfy the requirements and specifications but **ease the process of building / producing what was designed**
 - Also to design items needed for production, operation and maintenance of what was produced through the project
 - **End-items:** Items from the main contract (or **product being build or produced**)
 - **Side-items:** Items items which need to be **produced to support the main items, or end-items**

5.3 Production / Build Stage



- Design now completed, time for:
 - Production / Manufacture / Construction / Fabrication
- Activities in the production / build stage:
 - System fabrication, testing, and planning for implementation
- System Fabrication: Actually doing the work (or overseeing the work), the project manager has the following responsibilities:
 - Monitoring, inspecting, documenting progress, comparing planned versus actual, and taking corrective actions

5.3 Production / Build Stage



- Planning for implementation:
 - This entail handing over or installing what was constructed, or produced to the user
 - Prime activities, **installing** and **training** the user
 - We need to plan for this to make sure the user gets the product, and can use and maintain the product, and that the new product is incorporated into the organisation without any serious disruptions to the existing environment and systems

13.3 Monitoring and Control Process



- Monitoring and control is the process of keeping the project moving as outlined in the execution plan
 - Project Control: Assessing progress against planned objectives / performance and taking corrective action
 - Process of Control and Monitor:
 - Setting performance standard
 - Comparing actual performance to standard
 - Taking corrective action to remove variance

13.3 Monitoring and Control Process



- Project Monitoring
 - Tracking the project
 - Assessing how the project is doing
 - Forecasting the future of the project
- Project monitoring involves collecting various project data to make informed decisions about the health of the project

13.3 Monitoring and Control Process



Internal and External Monitoring and Control

- Internal control – contractor, or party constructing / producing the product
- External control – control imposed from a client, user or other regulatory party or stakeholder

13.4 Project Monitoring and Control Emphasis



- Scope creep: Scope's tendency to grow over time (this must be controlled)

Scope Control

- Aim of scope control:
 - Where requirements or work change occurring
 - Ensuring these changes are necessary and beneficial
 - Restrain changes as far as possible
 - Manage the implementation for changes

13.4 Project Monitoring and Control Emphasis



Quality Control

- Managing the work to achieve desired requirements and specifications
- Addressed in the quality management plan

Schedule Control

- Keeping the project on schedule
- Apply buffer strategically to ensure realistic completion dates and times

13.4 Project Monitoring and Control Emphasis



Schedule Control

- Fever chart is a graphical representation of how the project is doing, in this case in terms of its schedule, showing how bad, or good it is going (red, green, yellow)
- We protect our critical path (or critical activities) by communicating the status preceding activities on the critical path therefore know the impact your activity is going to have on the project

13.4 Project Monitoring and Control Emphasis



Procurement Control

- Controlling the output of the project by controlling the suppliers
 - Inspections and suppliers' capability assessment
- All according to the Project Procurement Plan

Cost Control

- Tracing variances in expenditures versus budgets and cost plans
- Eliminate unauthorized and inappropriate expenditures
- Minimize and contain cost changes

13.5 Work Packages and Control Accounts



- Work packages and control account are key elements of project control
- Control account consist of work packages
- Work package is like a contract for a specific activity or job
 - Its own requirements
 - Work description
 - Budget
 - Cost
 - Schedule
- Therefore, a key point of data collection is to control the project
 - We manage on an activity / work package / control account level

13.5 Work Packages and Control Accounts



- Work authorization
 - Part of project control
 - Work authorization / start-stop control: Work starts only after formal authorisation and is stopped only after review and acceptance
- The same applies on phase and project level as a whole

Thank you
Enkosi
Dankie



Photo by Stefan Els